

Project Amazon: An Open Class 60 Sailboat for Single-Handed Round-the-World Racing

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Project Amazon is a custom-designed Open Class 60 sailboat that began development in 1992. Detail design took place in 1995-96, and construction occurred in 1996-97. *Project Amazon* is an aluminum cat-ketch with two carbon fiber, free-standing, rotating wingmasts, an aluminum asymmetrical swinging keel, and twin carbon fiber, daggerboard-style rudders. Followers of the Around Alone and Vendée Globe single-handed, around-the-world races and developments of Open Class 60 sailboats may find these features unusual because collectively they do not fit mainstream racing sailboat design. None of *Project Amazon's* design features is totally new to sailboat naval architecture. These ideas have all been tried before, and all have been successful. But this is probably the first time that so many innovations have been combined into a single major racing sailboat design.

Introduction

ANY VESSEL DESIGN, whether a racing sailboat such as this, or a cargo ship, begins life from a list of owner's require-

ments. *Project Amazon's* owner, Sebastian Reidl (RI-dle, rhymes with bridle), had requirements much the same as all round-the-world single-handed racing skippers: the boat must be fast, strong, reliable, and easy to sail in a wide variety of conditions, from dead calm in the doldrums to gales with 100 ft (30 m) waves in the Southern Ocean. This sailboat is to be a machine, not a fancy yacht. It must be easy to fix should anything break, which is bound to happen somewhere, sometime. Everything in the boat is designed to mini-

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mize the chances of structural failure without sacrificing speed or the safety of the skipper and the sailboat.

Primary features

Hull design

- Dory-type hullform with an aft skeg
- Lifting strakes full length of hull
- Upper spray rail
- Air lubrication system
- Aluminum construction

Appendages

- Asymmetric keel shape with faired trim tab
- Aluminum keel construction
- Fuel tank inside keel
- Swinging keel
- Twin daggerboard rudders in turning cylinders
- Carbon fiber rudder construction

Rig

- High aspect ratio, elliptically-shaped mainsails in free-standing, cat-ketch arrangement
- Rotating wingmasts
- Carbon fiber mast construction

Hull design

General

Project Amazon has a considerably different hull shape from other Open Class 60 designs. It has a shallow, low-deadrise hull with a skeg fitted at the aft end (see Fig. 1). The stern is narrow aft rather than "aircraft carrier" wide as is typical of most Open Class 60 designs. Lifting strakes run full-length along both sides of the hull. A frigate-type spray rail is built into the upper topsides forward. Finally, the hull bottom has air-lubrication vents to minimize drag. Except for one other instance of air lubrication vents, none of these features are found on typical round-bottomed Open Class 60s, although they are not new to yacht design.

Project Amazon is built in aluminum. Open Class 60s have been built in aluminum before, notably Philippe Jeantot's first two *Credit Agricoles*, Jean Luc Van Den Heede's *36.15 MET*, and Alain Gautier's *Generali Concorde*. But in the last

few years, more and more very lightweight composite hulls are being built.

Dory-type hullform with an aft skeg

The low-deadrise bottom shape holds course better and is easier to steer than round-bottomed hulls. Round-bottomed hulls have minimal wetted surface which theoretically means less friction drag and faster speed. But their "skimming dish" shape makes them hard to control, particularly in large waves such as in the Southern Ocean. Hulls very wide aft have tremendous stability which makes them stand straighter to the wind, which is fine in relatively light conditions. In heavier conditions, however, as wide boats heel more, they become hard to control because the rig forces do not balance very well with the hull and keel forces.

Project Amazon's hull has minimally more wetted surface than its competitors, so friction drag may be a touch greater. This is compensated for by having much more sail area in the cat-ketch rig over the more common single-masted, stayed rig. Power-to-weight ratio is relatively high. Also, the hull is relatively wide, but only in the middle. The aft end of the hull is narrow so that it balances better with the shape of the bow. As the boat heels, the forces between the rig and hull balance better naturally. As a result, the hull is more easily driven, requires less effort to steer, and is much easier on the skipper and the autopilots. Also, it will not broach as readily, so it will be safer in heavy seas.

The aft skeg was incorporated into the hull shape to improve directional stability. It is plated on port and starboard sides and was built to be easily cut off should it be found unnecessary. Ultimately, the owner cut off the skeg prior to the boat being launched, for reasons of reducing weight and wetted surface, as it was becoming apparent that *Project Amazon* was getting heavier than expected during construction. Reports from the owner indicate that *Project Amazon* has more than adequate directional stability without the skeg.

Lifting strakes full length of hull

Although most Open Class 60s are able to plane like a powerboat, the common round-bottom hull is not a good planing shape. The bow buries into waves easily, and the boat is hard to control. At Sponberg Yacht Design Inc. (SYDI), we

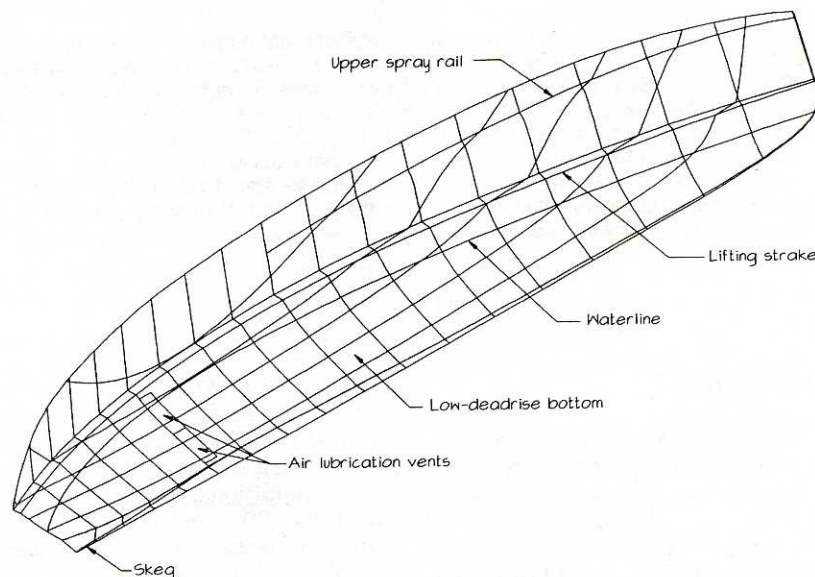


Fig. 1 Perspective of hull lines of *Project Amazon*

have analyzed photographs and videos of Open Class 60s to study how their bows submerge into the waves, which results in a lot of extra drag and slows the boats down. Powerboats have had lifting strakes for the greater part of this century; they lift the hull over the waves, thereby keeping wave drag to a minimum. *Project Amazon*'s lifting strakes are very similar to those on high-speed powercraft, but tailored to the requirements of a sailing hull. They also provide a great increase in stability as they submerge when the boat heels, so even at low speeds they have a beneficial effect in keeping the hull upright and fast.

Upper spray rail

Some waves are just too big to sail over, and so the boat goes through them. The hull has a pronounced flair in the topsides forward, finishing in a knuckle, which throws water out and away from the hull rather than letting it wash over the deck. This feature is common to the bows of high-speed military frigates.

Air lubrication system

The air lubrication system minimizes friction drag by allowing air to feed into the water underneath the hull. This feature works only when the boat is planing.

As originally designed, there was to be a series of holes drilled through the hull plating between the keel and the lifting strake, positioned about $\frac{3}{4}$ of the hull length back from the stern. The correct fore/aft position of the vent holes and the initial design concept was obtained by consultation with Lars Bergstrom of B&R Designs in Sarasota, Florida. Final construction details were developed by SYDI (see Fig. 2).

Flush-fitting ultra high molecular weight (UHMW) plastic cover plates fit over the vent holes to keep them covered at non-planing speeds. The holes opened into watertight manifolds built between the bottom floors and frames. Attached to these manifolds were pipes, fitted with seacocks, leading to the cockpit sole. Two manifolds were to be fitted on each side of the hull with one pipe to the cockpit for each manifold, for a total of four manifolds and four cockpit air vent pipes.

At the appropriate speed, the vent covers in the hull bottom are opened by hand-actuated levers in the cockpit, and air is drawn from the cockpit to underneath the hull by natural suction of water flowing past the aft open edges of the vent hole covers. The effect of the air vents is to maintain high speed in the planing regime for longer periods. When *Project Amazon* is not planing, the air vent covers are closed flush to the hull.

Hunter's Child, a Lars Bergstrom-designed Open Class 60 that sailed in the 1994 BOC Challenge (now called the

Around Alone), also had an air lubrication system. *Hunter's Child*, however, had a round hullform, not a deadrise hullform. As a result, the covers over the vent holes had to be fixed in the open position which created extra drag at non-planing speeds. Opening/closing covers would have been too expensive and complicated to build onto her round-bottomed hull. In this respect, *Project Amazon*'s deadrise hullform is better suited to an opening/closing air vent system by virtue of its relatively flat bottom plating.

Unfortunately, the owner was trying to cut back on costs throughout the latter part of the boat's construction, and so the air lubrication system was built without the opening/closing covers. Instead, the vent holes were drilled into the hull at about a 45-deg angle, sloping up and forward, and so were left perpetually open (Figs. 3 and 4). Reports from the owner after his transatlantic crossing from Cape Town, South Africa, where *Project Amazon* was built, to Puerto Rico, were generally positive. Although the effectiveness of the air lubrication system has never been properly and fully measured, the owner said that he can definitely hear and feel the air being drawn from the scuppers in the cockpit, and he thinks that the boat holds itself in the planing regime for long periods of time as a result.

Aluminum construction

Most Open Class 60s are built of composite materials such as fiberglass and carbon fiber because they are very lightweight. But composites are brittle materials, and when they hit something hard enough like flotsam or ice, they break. When they do, the boats frequently sink. If the boats survive, their composite hulls need a warm, clean, dry environment for repairs. Where is such a place in the middle of the ocean?

Aluminum dents. It is a ductile material and very difficult to break. The likelihood of catastrophic failure and sinking are much less than with composite hulls. And aluminum can be welded anywhere, even at sea. Overall, the durability, safety, and ease of repair are much better than on a composite hull. To compensate for its extra weight, extra sail area is required to drive the boat. The cat-ketch rig on *Project Amazon* allows a very favorable amount of sail area in proportion to hull weight than do the conventional rigs of its competitors. So the weight of the aluminum is not necessarily a disadvantage.

Project Amazon is transversely framed rather than longitudinally framed at the request of the owner, who is a machinist by trade and had built transversely framed aluminum boats before. Each of the frames and bulkheads comprises flat panels, or panels with just one degree of curvature, so that their shapes lend themselves quite well to angle-bar framing. Specifications for the framing and plating are

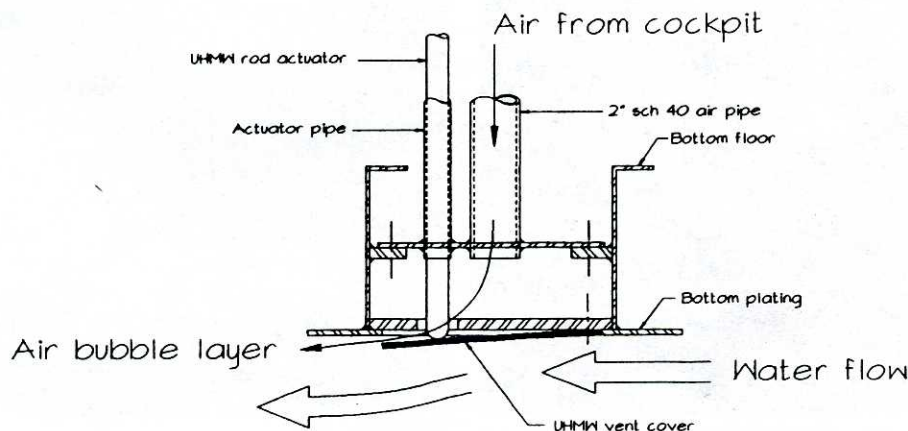


Fig. 2 Original detail of air lubrication vent through hull

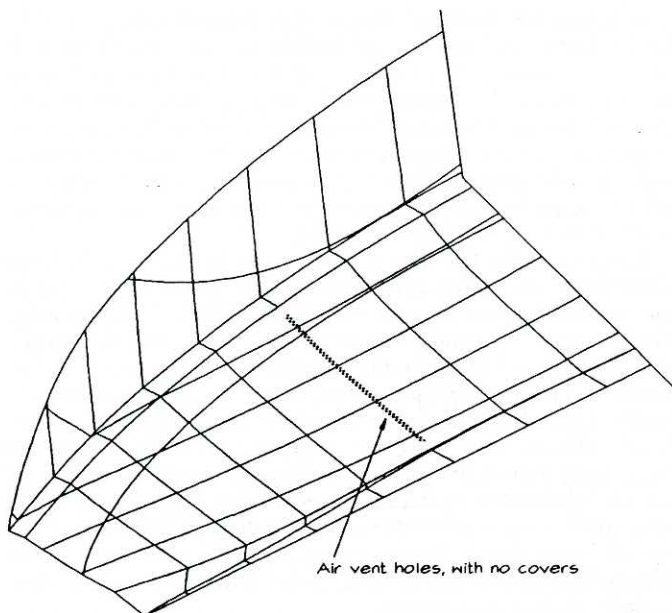


Fig. 3 Revised air vent hole arrangement

shown in Fig. 5. Photographs of the framing under construction, and of the plated hull at the bow and the stern are shown in Figs. 6, 7, and 8.

Appendages

Asymmetric keel shape with faired trim tab

Asymmetry refers to the cross-sectional shape of the keel blade. Most sailboats have symmetric keels because they must sail on opposite tacks. But asymmetric keels have much

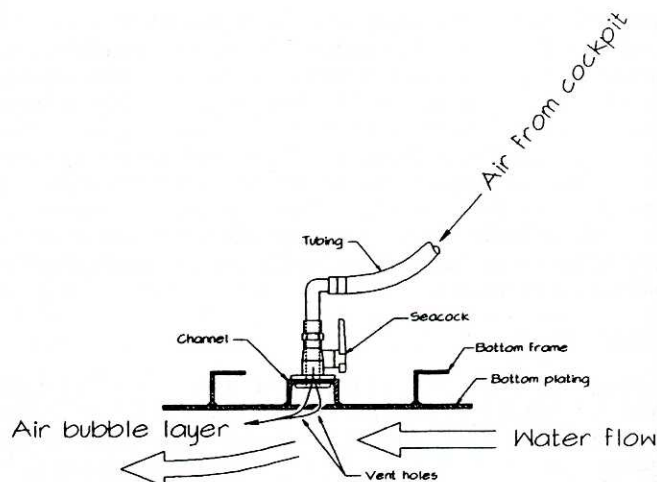
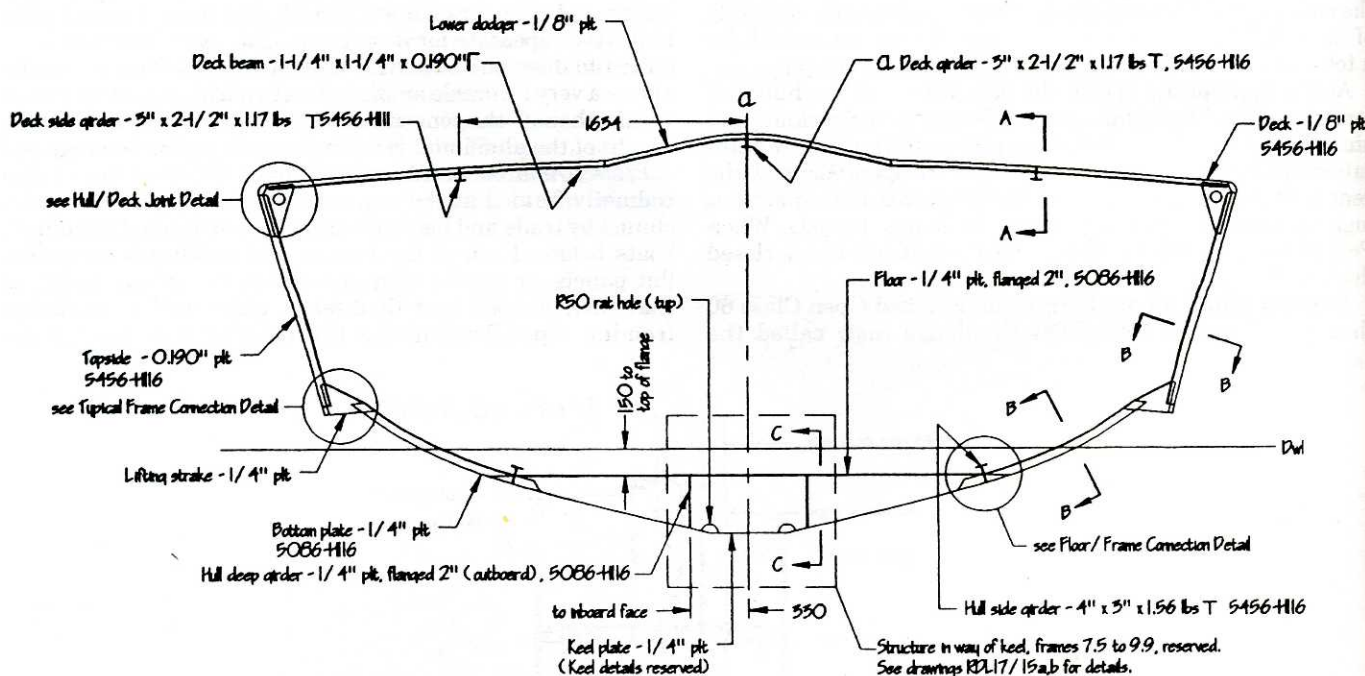


Fig. 4 Revised air vent construction detail

higher lift at low angles of attack, so that they make the boat point towards the wind much better. This is accomplished by the design of the trim tab on the back of the keel blade (Figs. 9 and 10).

The aerofoil section used for the keel on *Project Amazon* is a modified GA(W) section of 21% thickness. GA(W) stands for General Aviation (Whitcomb), after the person who designed the section for general aviation aircraft. GA(W) sections are scaleable so that one can change their relative thickness by a simple scale factor.

Most designers use 9% to 15% thickness sections for sailboat keels in order to reduce profile drag to a minimum. Taken to the extreme, thinner section keels tend to be more difficult to attach to the hull. Also, thinner sections bend and twist more than fatter sections, and this can cause hydrody-



Midship Section (Frame 9.0, view fwd)
(scale 1:20)

Fig. 5 Midship construction section (excerpt from construction plans, scale reduced)

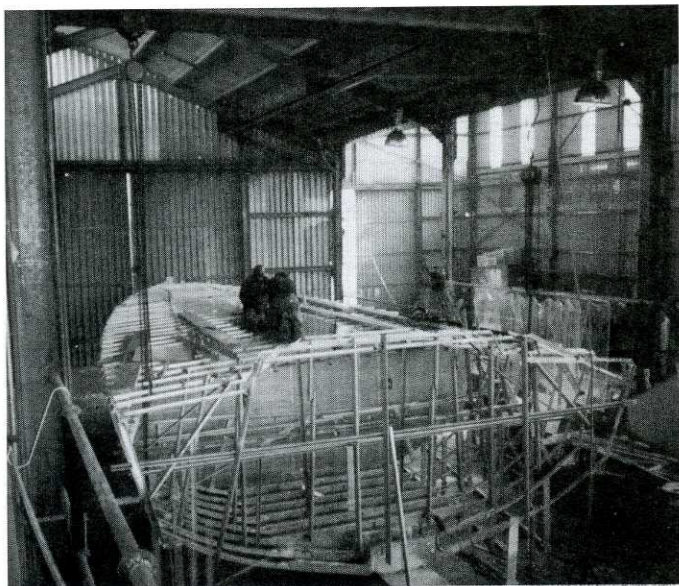


Fig. 6 Project Amazon's transverse framing

namic imbalances if they are not structurally stiff enough. Finally, thinner sections have what can be referred to as a narrow open window of acceptable angle of attack to the flow before separation off the leading edge (stall) occurs. When the

loads on the boat are relatively light, such as racing in calm water in light air, then these characteristics of thin sections are of minor consequence. However, once the boat starts sailing in waves, narrow keels can be very ineffective, even dangerous.

A fatter aerofoil section has several advantages over a thinner one. First, the extra drag associated with a fatter section is negligible compared to a thinner section. Inspection of profile drag coefficients reveals that thick and thin sections have very similar drag coefficients in both laminar and turbulent flow. Indeed, in laminar flow on some sections there occurs a "drag bucket," that is, a region of exceptionally low drag coefficient at low angles of attack (Figs. 11 and 12). Thin aerofoil sections have a very narrow drag bucket, and thicker sections have a relatively wide drag bucket. What this means is that in waves, when the motion of the hull causes the angle of attack to the keel to change greatly, thin aerofoil sections tend to stall more readily than thick sections. A thicker section will keep flow attached better in waves, and it will experience less speed loss due to stalling. This more than makes up for any loss of speed due to the thicker overall dimension of the keel.

Aluminum keel construction; fuel tank inside keel

Another advantage of a thicker aerofoil section is the obvious increased moment of inertia and section modulus of the section for strength and stiffness purposes. Thicker sections

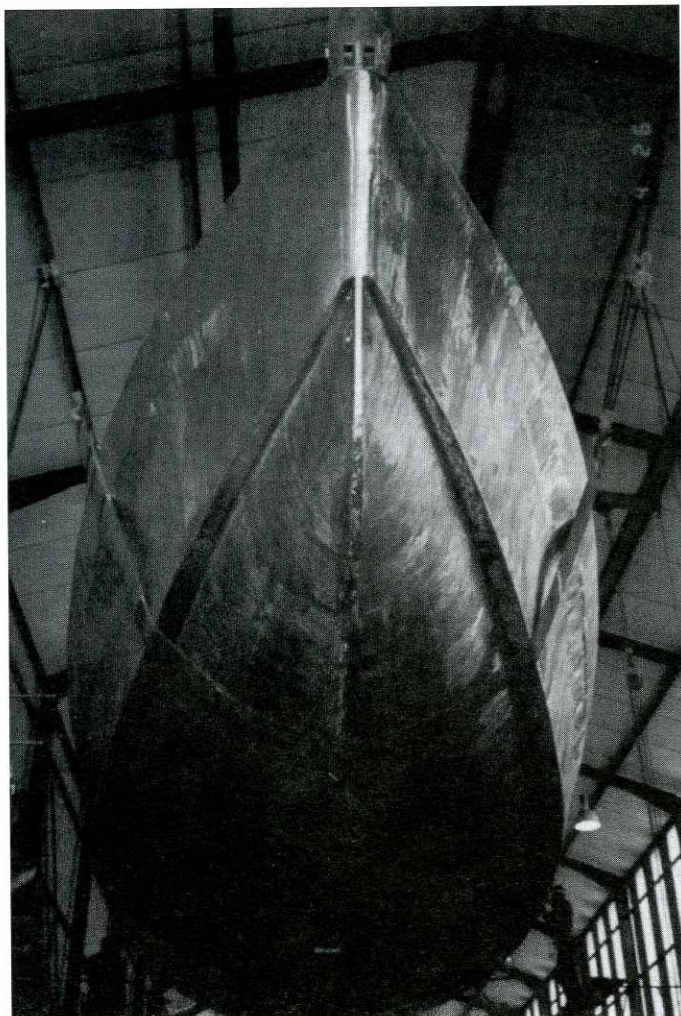


Fig. 7 View of lifting strakes at bow. Forward ends of strakes were filled in with an epoxy radius

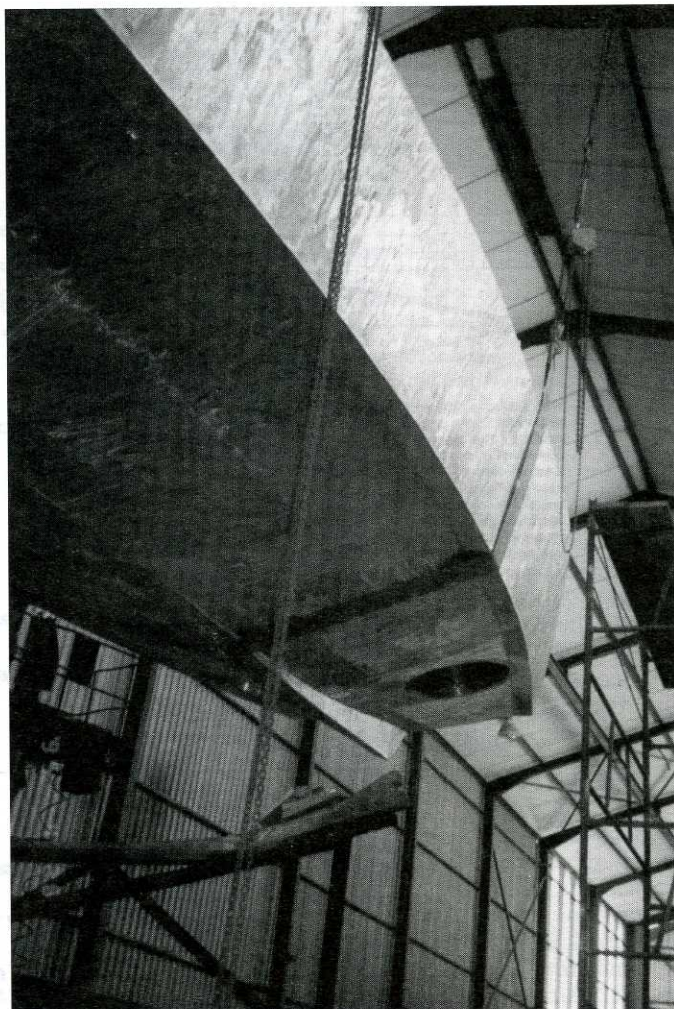


Fig. 8 Aft end of hull showing lifting strake, rudder cylinder opening, and skeg. Skeg was removed before launching

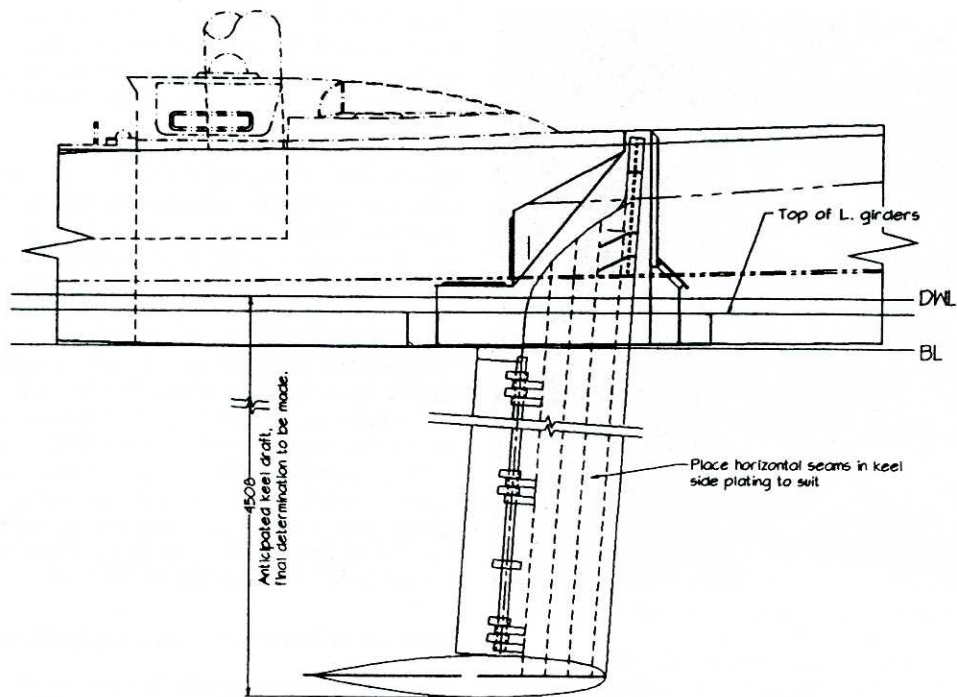


Fig. 9 Keel profile. Keel is a hollow aluminum structure made primarily of 1/2-in.-thick (12.7 mm) plating. The inside of the keel blade is a diesel fuel tank for powering all on-board power systems

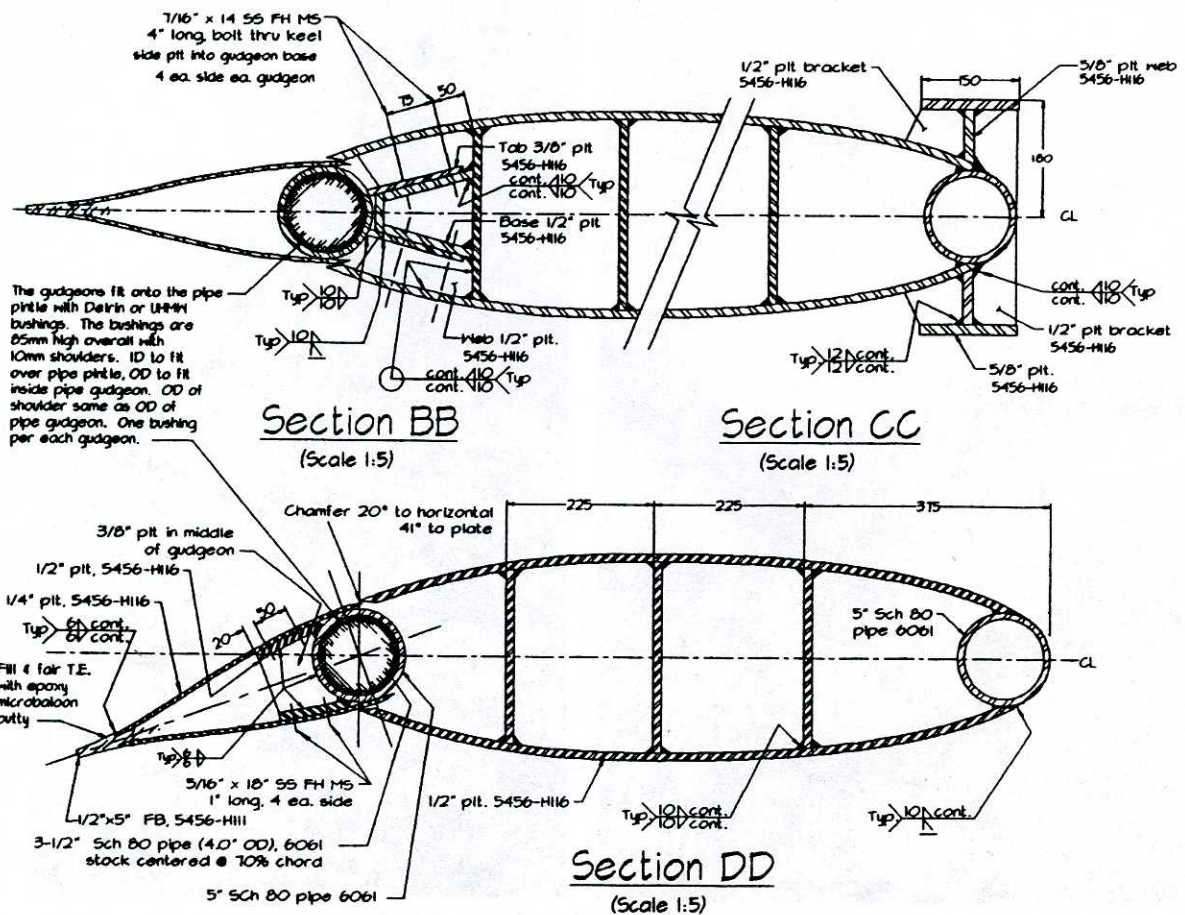


Fig. 10 Sections through keel showing trim tab centered (above) and deflected 20 deg (below)

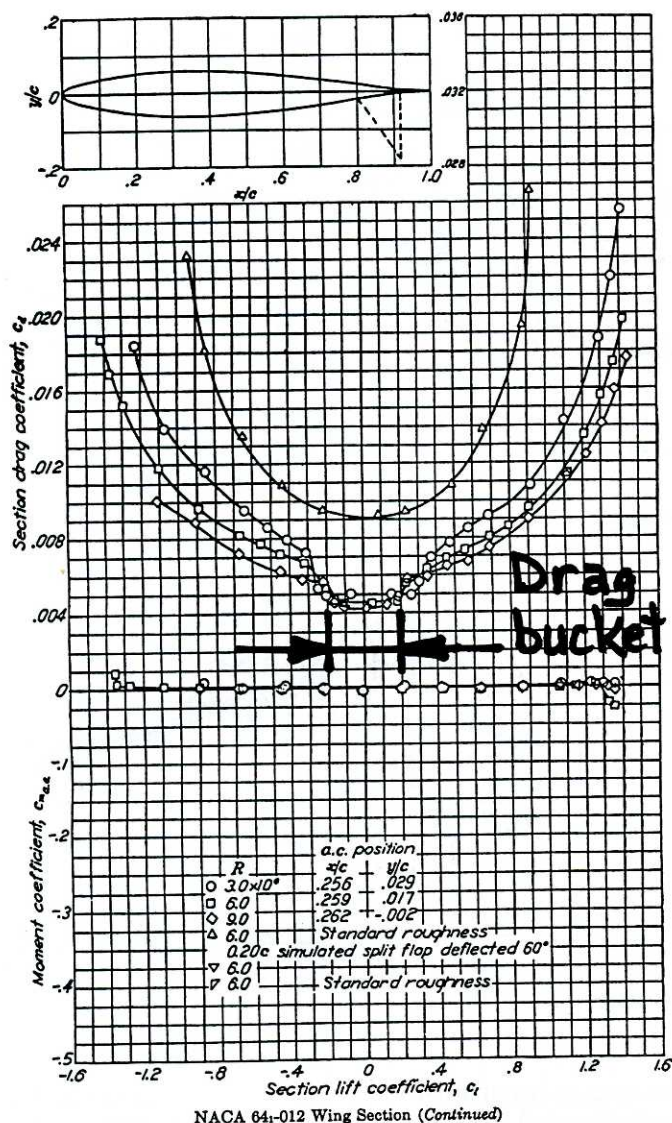


Fig. 11 Lift/drag curve for a typical thin aerofoil section. Note narrow drag bucket

will not bend and twist as much as thinner sections, so structural reliability is better and easier to accomplish. A stiffer keel also has more uniform and reliable hydrodynamics; that is, fewer anomalies of flow will occur due to insufficient stiffness of the keel blade. In *Project Amazon's* case, the owner wanted an aluminum keel blade rather than a steel or carbon one, and the only way to achieve sufficient strength and stiffness in the keel structure was to utilize a relatively fat aerofoil section. The 21% aerofoil section allowed the use of 1/2-in.-thick (12.7 mm) aluminum plating to create a hollow keel structure, and this, in turn, allowed the use of the space inside the keel blade as a fuel tank. Fuel has to be carried anyway, and better for the boat's stability that it be carried low in the keel rather than in the hull double-bottom tanks. The nearly one tonne of fuel carried inside the keel blade is estimated to be sufficient to power the diesel-electric generators on a voyage nearly twice around the world.

Finally, incorporating a faired trim tab into the keel blade—that is, a trim tab which, when deflected, has a completely fair surface from leading to trailing edge on the lift (low pressure) side—is more easily accomplished on a thicker section than on a thinner one. The geometrical fairing is simply easier to do. In *Project Amazon's* keel, the trim tab comprises the after 30% of the aerofoil section, and the trim tab

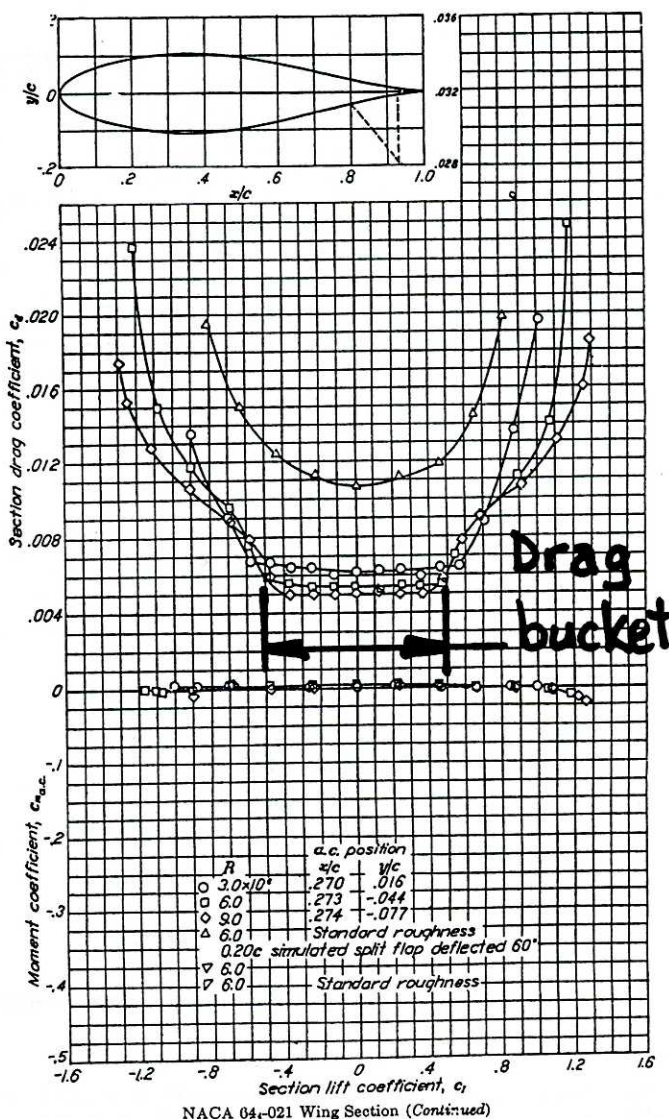


Fig. 12 Lift/drag curve for a thicker aerofoil section showing wider drag bucket. When an aerofoil shape is made asymmetric, the drag bucket stays the same width but shifts to the right

can deflect up to 20 deg, giving an overall foil section angle of attack of 6 deg. With the faired trim tab deflected, the drag bucket shifts in a favorable direction, and lift increases dramatically. That is, you get high lift with very low drag. Therefore, depending on the conditions at hand, the skipper can set the trim tab trailing edge to leeward to create more lift that will carry the boat to windward, or he can set the trim tab trailing edge to windward to counteract heeling moment. In other words, the skipper can arbitrarily establish high lift on either side of the keel at will in any sailing condition.

Swinging keel

Besides asymmetry, the keel can swing up to 30 deg to port and starboard in a matter of seconds to increase the boat's stability on either side, as necessary. This helps the boat sit more upright which is better for low drag and high speed. Many Open Class 60s rely on many tons of water ballast for added stability which takes some minutes to transfer from side to side. So while their at-rest low weight may look attractive at first, most of the time their water ballast brings their overall weight much closer to *Project Amazon's*.

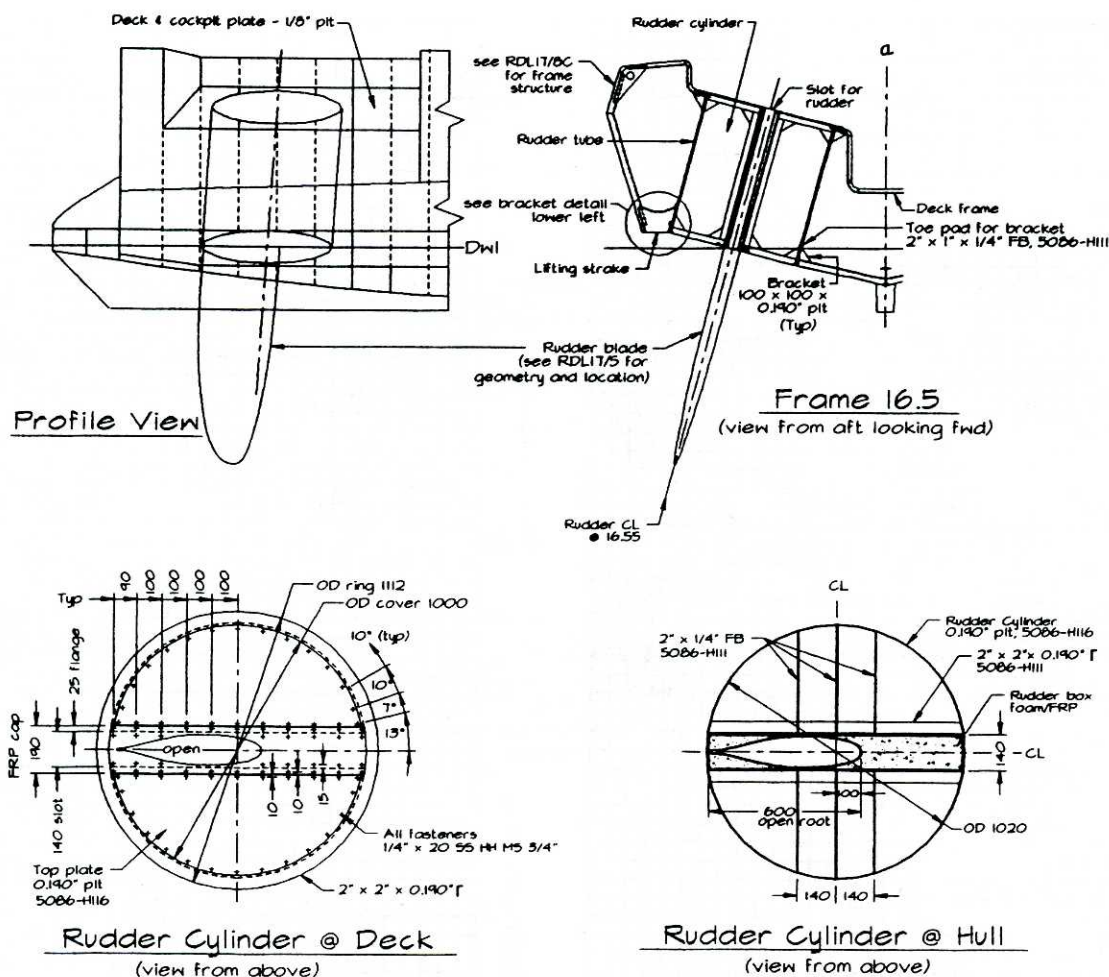


Fig. 13 Rudder general arrangement and basic cylinder construction

Most boats with swinging keels suffer greatly from the loss of lateral plane in the underwater profile as the keel swings to windward. This tendency is aggravated by the wide, round, shallow hullforms typical of the type and which have very little grip with the water. These boats tend to slip to leeward quite excessively, and so their speed made good to windward is nothing to write home about. To compensate, boats with swinging keels usually have a pair of asymmetric daggerboards situated forward in the hull port and starboard and projecting down and outward about 15 deg. On upwind legs, the leeward-side daggerboard is deployed, and its asymmetric shape and slight toe-in help pull the bow back to windward. Of course, these are other pieces of equipment that must be built, maintained, and repaired if broken, which is quite likely as they are in a very vulnerable position in the front of the boat. Forward daggerboards also create additional drag.

Project Amazon's low deadrise, dory-type hullform with full-length lifting strakes has natural directional stability. Also, the LCB moves very little from upright to heeled position (unlike the wide-aft hullforms common to Open Class 60s), so she has a natural tendency to hold her bow high and point to windward. She holds course well, and so forward daggerboards are not required.

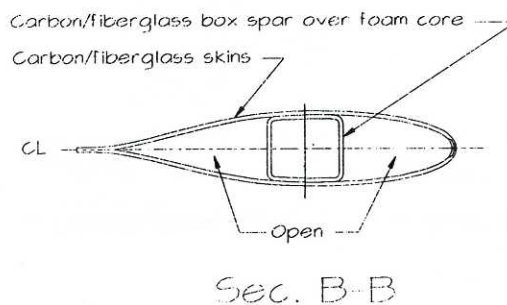
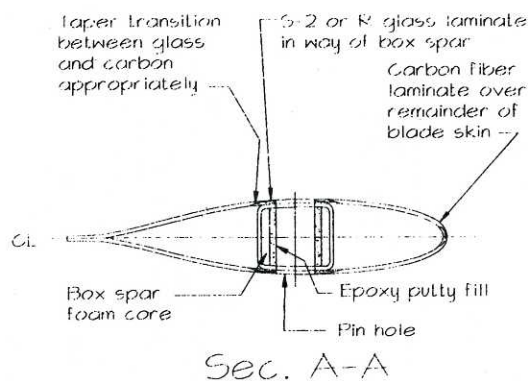
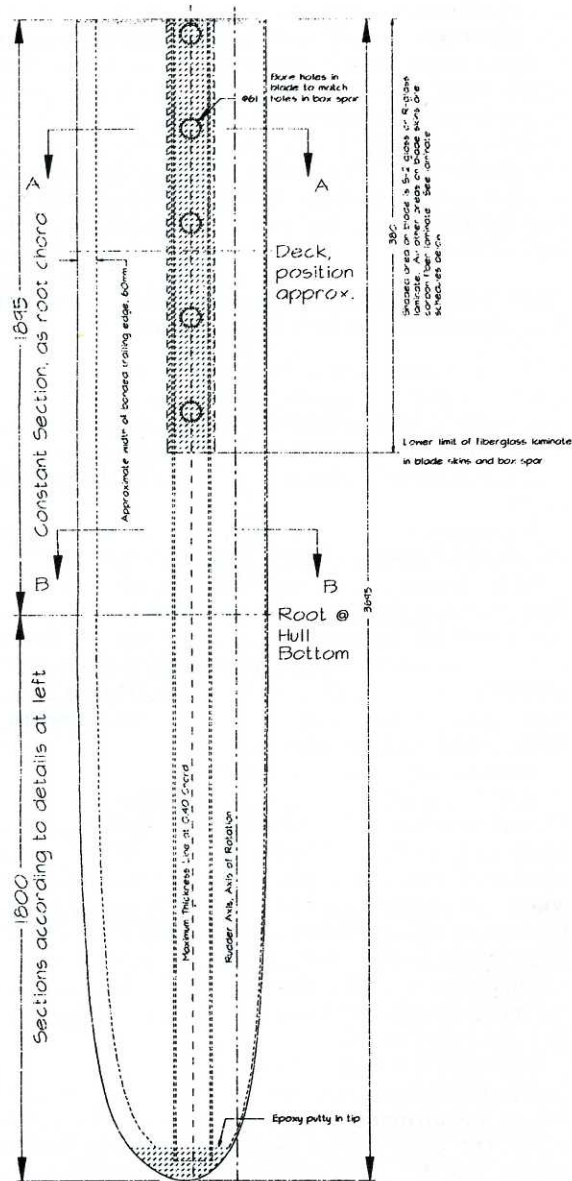
Twin daggerboard rudders in turning cylinders

A recent development in yacht design is the mounting of the rudder blade inside a turning cylinder. *Project Amazon*

has such an installation with two rudders offset from the hull centerline and canted outboard 15 deg. Twin rudders offer twice the steering reliability of single rudders, and they operate in the cleaner water flow outside the downwash of the keel. As daggerboards, they do not have rudder stocks which are difficult to build and keep sealed from the sea. The daggerboard rudders can be set to any of five pre-set depths to vary steering load depending on sailing conditions. They can be partially or fully withdrawn as necessary to keep drag low, consistent with controllability. If the rudders hit something, they break cleanly without damage to the hull or steering systems. They would be easily replaced with on-board spares.

As designed, the rudders, are a 15% GA(W) aerofoil section with a 600 mm (24 in.) chord length (Figs. 13, 14, and 15). They are built of carbon fiber and are housed inside a fiberglass box casing which itself is situated inside a closed aluminum cylinder. The steering center of the rudder blade, just forward of the 1/4-chord line, is situated on the cylinder axis so that the rudders balance well. The reason for using a central fiberglass box casing is so that if the shape or balance of the rudders need to be changed to improve steering performance or balance, then just the rudder blades themselves, and the fiberglass casing if necessary, would be changed, and not the entire aluminum cylinder assembly together with its bearings.

One steering ram from each of the twin hydraulic steering systems is connected to each rudder cylinder, and the cylinders are mechanically cross-connected with a tie-rod. Two



RUDDER CONSTRUCTION

Fig. 14 Rudder construction details, hollow-molded carbon fiber and fiberglass laminates

autopilots are also installed in the steering system, one to each cylinder. In case any one rudder, hydraulic steering ram, or autopilot should fail, the other of the pair of each can take over. There is also a mechanical linkage direct from the steering wheel that can be engaged/disengaged in a matter of seconds without any tools should there be a total system failure. Under normal conditions during an around-the-world race, steering is accomplished by joystick direct to the autopilots, or by computer control from on-board GPS navigation direct to the autopilots.



Fig. 15 Author being interviewed by newspaper reporter on board *Project Amazon*. Note the starboard-side rudder with the lifting handle on top of the blade, the lock pin, and the hydraulic tiller arm connected to the cross-connecting mechanical linkage



L_{oa}	18.280 m
L_{wl}	18.153 m
B	5.728 m
D	4.500 m
Displ.	14.915MT(to D_{wl})
Sail areas:	
main	125.299 m ²
mizzen	125.749 m ²
Small headsail	95.860 m ²
Small staysail	95.229 m ²
Large headsail	126.563 m ²
Large staysail	126.834 m ²

Fig. 16 Sailplan of Project Amazon. Original design particulars are given above

neering committee that he could successfully steer the boat with the trim tab on the back of the keel. The "scrutineering" committee accepted the keel trim tab as the emergency steering system.

Rig

General

The most noteworthy feature of *Project Amazon* is the rig (Fig. 16). It is a cat-ketch, which means it has two masts: one mast in the bow (the "cat" part), and a second mast between the rudder and the keel (the "ketch" part). Two masts can fly more sail area than one mast. Also, these masts don't have any wires holding them up as most sailboats do; they are free-standing. The masts are made out of carbon fiber which is incredibly strong and lightweight enough to not require wires for support. Finally, the masts are shaped like wings which rotate into the wind for the cleanest airflow on all points of sail. Most sailboat masts, by virtue of their stays and shrouds, are fixed and cannot rotate. The airflow around such rigs inherently has a lot of drag and not very much lift.

Never before in the history of offshore yacht racing has a

monohull ever competed in a major ocean event with such a rig. However, none of these ideas is new; for example, masts without wires to hold them up are as old as the first tree trunk stuck in a raft. The last and closest design circumstance in recent times was *Lady Pepperell*—a 54 ft cat-ketch with free-standing masts that rotated, but were not shaped like wings—that raced in the inaugural 1982–83 BOC single-handed race around the world. Unfortunately, she lost her keel on the second leg and now lies on the bottom of the Indian Ocean.

High-aspect-ratio, elliptically shaped mainsails in free-standing, cat-ketch arrangement

Most modern offshore racing sailboats have just one mast which is fixed in place by shrouds and stays. The headstay and backstay define and confine the shape of the sailplan to a triangle. Some curvature in the mainsail roach stretches past the backstay a little, but basically the sails form a triangle.

A triangle is the absolute worst possible shape anyone could ever conceive of for a lifting surface.

Why? Because a triangle has inordinately high induced drag. Induced drag is that drag which is inescapably created along with lift. You can reduce it, but you can never get rid of it. In a triangular rig, the upper half of the sailplan creates almost pure drag with very little lift. This is because of the narrowness of the sailplan up high compared to its width down below, and the resulting airflow around this shape.

Theoretically, you can reduce a triangular rig's induced drag if you could do but one simple thing—twist the head of the sail to windward. Unfortunately, with fabric sails, this is impossible to do, and even if you could, twisting a sail to windward increases angle of attack preciously close to stall, where lift suddenly drops to zero. It would be a continual knife-edge balance between power and stall which cannot be done when you're sailing through waves—the rig moves around too much. So if we insist on wires to hold up the mast, we must accept a more or less triangular sailplan which is just not very powerful.

However, if we are bold enough to let the laws of physics govern rig design, then first we must do away with the wires, rather like airplane designers did back in the 1920s. Without wires, the sails are no longer defined and confined by triangles, but can be shaped elliptically which is much more aerodynamically efficient—least induced drag for most lift. Also, eliminating the rigging totally eliminates parasitic drag. And not only are the sails much more efficient, but larger ones and more of them can be raised. The two masts can carry a number of different kinds of large asymmetric spinnakers and staysails for generating great power and speed on all points of sail. The capabilities of the two-masted, free-standing, wingmasted cat-ketch rig, therefore, are tremendously greater than a single masted, conventional stayed rig.

Eliminating the wires in the rig also greatly improves safety. A stayed, triple-spreader rig such as on an Open Class 60 sloop has over 500 separate parts holding up the mast. If any one of those parts fails or falls out, the rig has, in the author's estimation, roughly a 95% chance of falling over the side. If that happens, and it does regularly in round-the-world races, the boat will most likely be out of the race. On *Project Amazon*, the two masts are held up by a total of 4 parts. Safety and reliability increase in proportion, 500+ to 4, roughly 12 500%! If it is not there, it cannot break.

Rotating wingmasts

The idea of a rotating mast is traceable back to at least L. Francis Herreshoff, who had a patent on a rotating mast

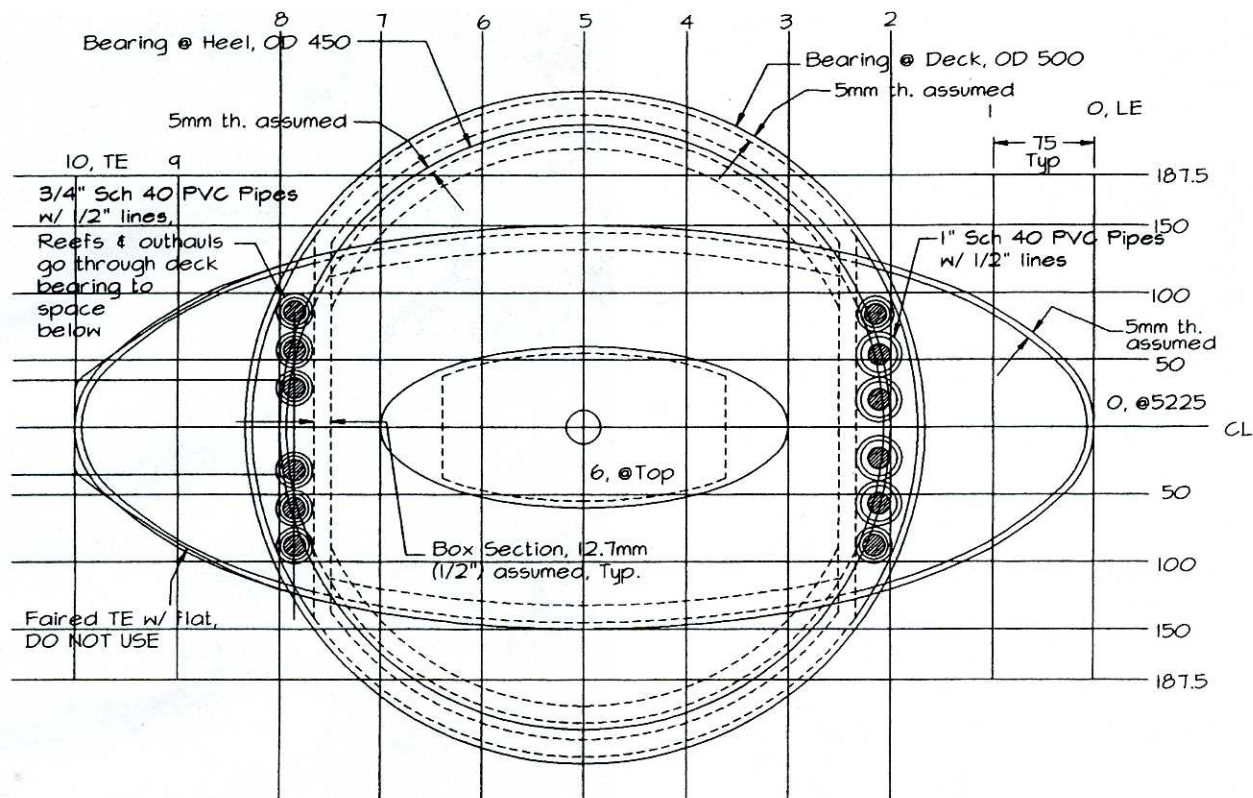


Fig. 17 Bird's eye view of Project Amazon's mast lines. Basic wing section is a 2.5/1 ellipse (750 mm chord, 300 mm overall thickness) at largest section

design, and he installed one on an R class boat ($L_{wl} = 20$ ft (6 m)) in 1925. But when the regatta committee of the New York Yacht Club heard about this rig, it promptly passed the rule prohibiting "revolving masts, double-luffed sails and similar contrivances." This prohibition remains. IMS rule 203.6 states, in its entirety: "Rotating masts are excluded for yachts measured under the Rule." No rule changes are in sight.

Even in spite of the rules, rotating a stayed mast is difficult to say the least because the rigging wires simply get in the way. Sailing on the wind, the airflow attaches fairly well over the mast and mainsail. Off the wind, however, the mainsail sets off the side of the mast, and its leading edge shape—the most important part of the rig for generating power—is awful. The airflow quickly separates off the mast. To recover some lift, the sailmaker has to build enough camber into the sail to fool the airflow into reattaching. Get rid of the wires, and full mast rotation is possible. Rotate the mast, and all the aerodynamics change. Even downwind—especially downwind—pure lift is much more powerful than pure drag. The boat is considerably faster because so much more power is harnessed from the wind.

Carbon fiber mast construction

The wingmasts are made of a prepreg carbon fiber box spar and prepreg fiberglass leading and trailing edge sections (Fig. 17). These are permanently bonded together with the box spar the primary structural element. On each mast, a stainless steel roller bearing is mounted at deck level, and a combination roller and thrust bearing is mounted at the heel. Both masts are very nearly identical so that they can be interchanged if necessary.

The basic aerofoil section for each mast is a 2.5/1 ellipse

(chord/thickness), for several reasons. First, an elliptical shape is very easy to engineer because cross-sectional geometric properties (area, moment of inertia, and section modulus) can be calculated by equations, which makes engineering the mast quite simple by use of a computer spreadsheet.

Second, the leading edge radius of an ellipse is exactly what airflow likes to see to remain smooth and attached to the surface. If the leading edge is too round, as in a circular section mast, airflow near the mast surface has to change direction too abruptly, and it can separate easily from the mast, thereby reducing lift and increasing drag. A round section is easy to engineer and build, but it is not the best shape aerodynamically.

Third, the symmetry of the entire ellipse makes for relatively easy construction because only one set of unique shapes need be made. On Project Amazon's masts, for example, one set of tooling was built for one side of the box section spar and another set for the leading/trailing edges. Therefore, for both masts, the two half-sets of tooling were cycled four times.

Fourth, experience has shown that the elliptical trailing edge provides a smooth transition of airflow from mast to sail with little or no eddies caught in the vicinity of the sail track and luff.

The ellipse should not be too narrow because narrower sections are more likely to buckle sideways. The mast should bend nicely both to the side as well as aft, and a fatter section does this well. Also, a fatter section has a wider window for angle of attack to the wind than a narrower section. When a boat sails over waves, the mast moves all over the place, and angle of attack on the rig, and thus lift, changes constantly and by huge amounts. If the angle of attack gets too high, the airflow separates off the leading edge, and the mast stalls. A narrow section stalls at a lower angle of attack than a fat section does.

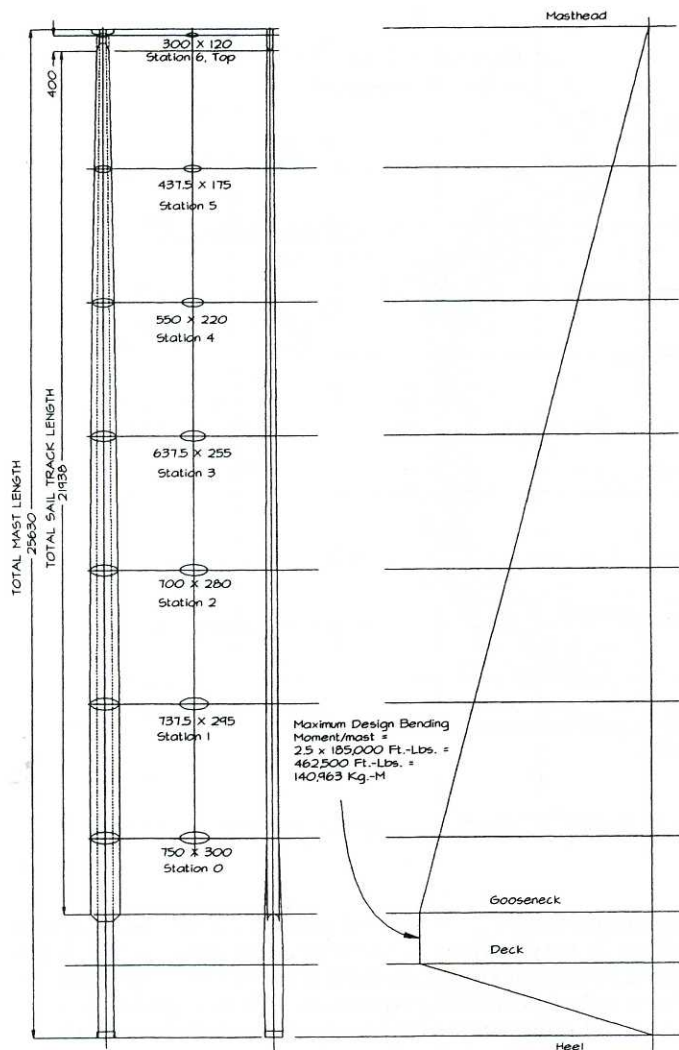


Fig. 18 Entasis taper in mast profile (at left), and design bending moment curve (at right)

Finally, a fatter mast is much more forgiving and easier to trim to the airflow than a narrower mast. *Project Amazon's* 2.5/1 ellipse is about as narrow as the author has gone on wingmast design. A ratio of 3.0/1 would be too narrow. A number of past designs are at 2.0/1, and quite a few production mast designs were done at 1.25/1, almost round.

Mast taper is also important, and *Project Amazon's* masts have a modified entasis (EN-ta-sis) taper, typical of SYDI's standard practice (Fig. 18). Entasis taper was used by the Greeks over 2500 years ago to build the columns of their temples to the gods—fatter in the middle and tapering non-linearly towards the ends—which prevents the illusion of concavity as produced by perfectly straight columns. Columns must also be stronger in the middle to prevent buckling, so entasis taper builds in strength.

Because a freestanding mast is basically a cantilever supported by the deck and hull structure, the strongest part of the mast, and, therefore, the largest cross-section, has to be at deck level. After a straight length from the deck up past the gooseneck, the mast tapers according to a mathematical formula, gently at first, then more so up toward the top. This modified entasis taper makes the mast fatter at mid-height than a straight taper would, and fatter is stronger, stiffer and lighter. When viewed from dock or deck level, the entasis



Fig. 19 *Project Amazon* racing in Antigua, April 1998

taper is more aesthetically attractive than a straight taper, the very reason the Greeks used it. At deck level and below, the mast has to be round for the bearings, and the bottom end has to pass through the deck opening. From deck to heel, a round section with straight taper is used for ease of construction and because aesthetic considerations there are not important.

The design of a freestanding mast is fairly simple. Whereas a stayed mast always carries its load in compression, due to the pull of the rigging wires, a free standing mast always bends. The pressure of the wind on the sails causes a bending moment load on the mast which is zero at the top, maximum at the deck and gooseneck, and zero again at the heel (Fig. 18). The maximum possible bending moment the mast can ever carry is equal to the maximum righting moment of the boat itself, simply due to the physics of a boat floating free on the water.

The design bending moment for the mast is simply the boat's righting moment multiplied by a factor of safety of 2.5. In composite poles that bend, the reinforcing fibers generally start to break at about 50% of the breaking load of the pole, causing permanent loss of strength. Therefore, setting the design breaking load of the mast at 2.5 times the maximum righting moment of the boat means that the actual bending load will never be higher than 40% of the design breaking load, and most sailing loads are much lower than this. Fibers won't break.

The mast must be as slender as possible consistent with strength, stiffness, and weight. Generally, larger cross-sectional shapes make lighter masts because the wall thickness gets thinner with increasing section size. There is a point where section size can be too big and wall-thickness too thin, making the mast too delicate to handle or subject to premature buckling. Therefore, every mast has an optimum size for each boat design, and wall thicknesses should be between $\frac{1}{8}$ and $\frac{1}{2}$ in. thick (3.2 to 12.7 mm) in most instances. Thickness-to-diameter ratio, t/OD , should be 0.03 or larger).

Sailing tactics

Other boats, and particularly Open Class 60s, rely upon very light weight to achieve exceptional broadreaching speed. But for at least half the race, that speed is in the wrong direction because to achieve those high speeds, those boats have to tack downwind at roughly 40 deg either side of the course line. Speed on the course line is actually 24% less than boat speed—15 knots on track becomes eleven and a half knots on the course. The conventional boats are also difficult to control because their round hull shapes are very shallow, and their exceedingly thin keels have minuscule projected area. As a result, they have poor directional stability and control. Speed without control is worthless. The propensity to crash is high.

Also, a boat with a stayed rig can sail perhaps 10 deg by the lee before the mainsail gibes. If not properly controlled, the boom will swing violently to the other side and crash against the lower shroud, perhaps breaking the shroud or itself. The boat may also broach if the seas are running fairly high. If something breaks or the boat broaches, it is no longer racing. In fact, it is instantaneously in danger of losing its rig and getting rolled over. It will be out of the race.

Project Amazon's rig gives it tactical advantages that other sailboats do not have. Since there are no wires, the boat is much more directionally stable and more resistant to gibing, a very important feature for around-the-world racing. The booms can set way forward of abeam wing-and-wing, and the boat can sail ninety degrees or more by the lee without gibing. If the boat starts to round up because of a hit by a wave or gust, the sails will naturally pull the bow back downwind. And even if the boat does gibe, what happens if the boom gets out of control? Nothing, because there is nothing to hit—neither wires nor boom can break. The sails will stop in a luff position all by themselves.

Also *Project Amazon's* hull design—the shallow deadrise, V-bottomed, dory hullform with the full-length lifting strakes—holds its course track exceedingly well, rather like a freight train on rails. *Project Amazon's* chances of broaching, and therefore, not racing, are exceedingly slim. That is why *Project Amazon* can do well in around-the-world racing: When the other boats crash, it just keeps on going.

Epilog

On September 26, 1998, owner/skipper Sebastian Reidl and *Project Amazon* departed Charleston, South Carolina on the first leg of the Around Alone single-handed race around the world. It was hoped that by the time this paper was presented, *Project Amazon* would be completing the race, and more material could be presented about her actual perfor-

mance. Unfortunately, three weeks after the start of the first leg, on October 15, Mr. Reidl retired from the race citing a water leak in his fuel supply in the keel. The problem was so severe that he could not keep up with cleaning fuel and racing the boat at the same time.

The contaminated fuel is probably related to a haul-out that occurred about a month before the race. *Project Amazon* was built about 4 tonnes heavy, roughly 30% heavier than intended. This is because very little weight management occurred during *Project Amazon's* construction. The owner declined to hire SYDI to monitor construction, and as a result, no weight data was available either from the owner or the shipyard. Indeed, SYDI was not able to participate in sea-trials when the boat was launched, and did not receive any freeboard measurements until the boat had been afloat for over a year.

Consultation with the owner during the summer of 1998 determined that the easiest way to shed some excess weight was to replace the ballast bulb with a lighter design. The owner was confident that the original bulb weighed 5 tonnes, and so a 3-tonne bulb would eliminate half of the excess weight.

With sketchy data from an inclining experiment that the owner executed by himself, a stability analysis showed that reducing the weight of the bulb by 2 tonnes reduced the range of stability from 141.4 deg to 133.9 deg, an acceptable change. The Around Alone race rules require all boats to have a minimum righting arm at 90 deg of 0.4 m. *Project Amazon's* 90 deg righting arm reduced from 1.9167 m to 1.5409 m with the bulb change, so it more than met the minimum 90 deg righting arm rule. The owner arranged for a new bulb design which was cast by Mars Metal in Ontario, Canada, and shipped to Palmer Johnson's yard in Savannah, Georgia, for the change that took place during August, 1998.

As originally designed, the keel blade was mortised into the lead bulb casting, and fastening bolts were oriented horizontally through the connection. To change the bulb, one had only to remove the fairing compound around the heads and nuts of the fastening bolts in the sides of the bulb, take them out, drop the bulb, set the new bulb in place, refasten the bolts, and fill and refair the bulb surface. During *Project Amazon's* construction, however, the owner authorized a change to the design which oriented the connecting bolts vertically. This meant the the bolts were no longer independent; rather they were studs fabricated into a cage that was cast inside the bulb, and the nuts for the studs were inside the keel blade. To unfasten the nuts to change the bulb, the side plating of the keel had to be cut open, and later, welded shut again.

It is very likely that the integrity of the fuel space inside the keel was breached when the plating was cut open to gain access to the nuts. This has not been confirmed, because to this date, now four months after her retirement (as this is being written), *Project Amazon* has not yet been taken out of the water for an inspection. It is known that the keel was not pressure tested after the bulb change. Also, *Project Amazon* had been afloat for over a year without any contaminated fuel incident prior to the bulb change, and the problem surfaced only afterwards. Until a complete inspection of the keel is made, this is the most plausible explanation for the contaminated fuel. It should be noted that this is the opinion of the designer/author only, without the benefit of inspection of the keel or consultation with the owner.

Mars Metal reported that the weight of the new bulb, as cast, was actually 3.286 tonnes, and that the original bulb weighed 4.598 tonnes when they received it back at their

facility. Therefore, the net change in weight was 1.312 tonnes, not the 2.0 tonnes hoped for. On the very morning of the race start, the author was able to take reliable free-board measurements that ultimately showed the weight of the boat, race ready complete with stores for an around-the-world race, was 19.475 tonnes, that her range of stability was 134.8 deg, and that her righting arm at 90 deg was 1.5224 m.

Project Amazon is now for sale, lying in St. Martin. It is

hoped that her new owners, whomever they may be, will allow the boat to be analyzed further so that her performance may be fully characterized.

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